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Evaluating Physics-Informed Learning for Fluid Dynamics

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Abstract

Physics-informed neural networks (PINNs) train on the governing equations instead of simulation data. We compare a PINN with a data-driven network and a finite-difference (FD) solver on 2D lid-driven cavity flow at Re = 100, measuring accuracy, physical consistency and computational cost on CPU and GPU.

Introduction

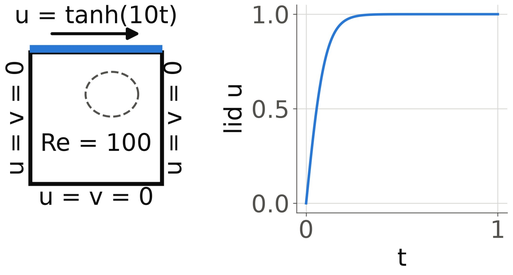
Does training with Navier–Stokes physics improve physical consistency enough to justify its extra cost?

RQ1 Accuracy of u, v, p against FD

RQ2 Continuity and momentum residuals

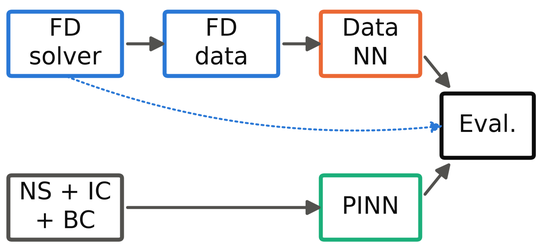
RQ3 Data NN vs PINN: consistency and repeatability

RQ4 Training time, inference time, memory, storage



Methods

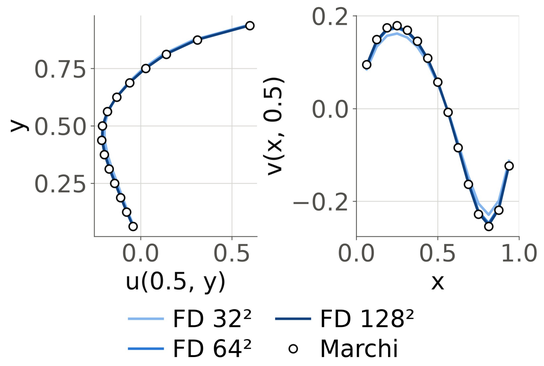
Unit cavity, lid u = tanh(10t). FD: 64×64 grid, Δt = 0.001, 51 snapshots t = 0–5. Both networks: (x, y, t) → (u, v, p), 5×64 tanh, 17,091 parameters, Adam 1e-3, 20,000 steps, 5 seeds. Data NN: FD data only. PINN: PDE residuals + initial and boundary conditions only (no FD data). Test A: 15% random held-out points; test B: 10 whole snapshots (t = 0.5, 1.0, …, 5.0) never seen in training.

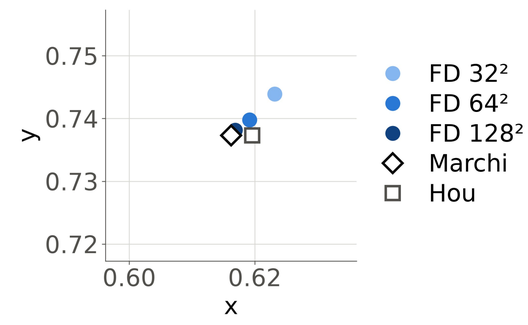


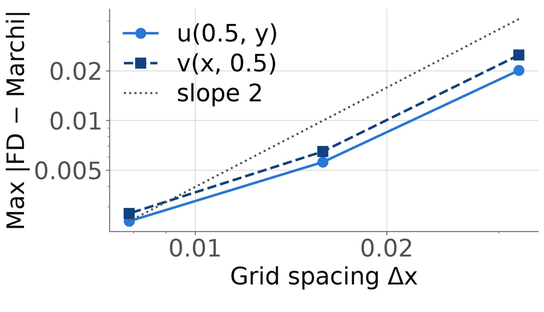
FD Validation

Steady FD solutions vs Marchi et al. (2009, 1024×1024): maximum centerline deviation of u falls from 0.0201 (32×32) to 0.0056 (64×64) and 0.0025 (128×128). At 64×64 the vortex center is 0.0039 from Marchi and 0.0025 from Hou et al. (1995), within one grid spacing (Δx = 1/63).

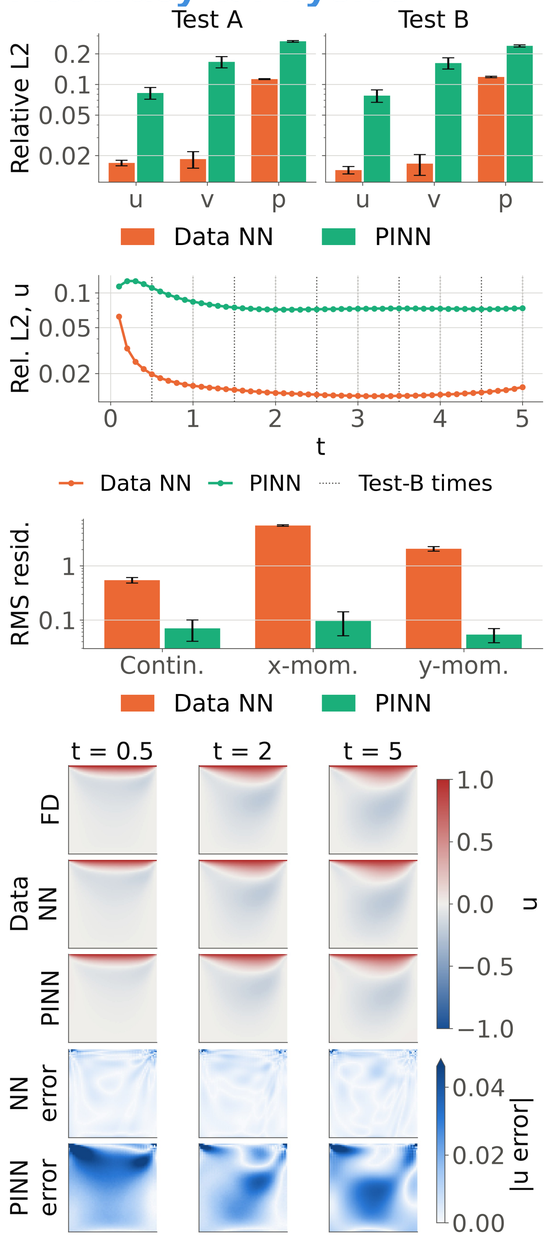
Transient check: halving Δt changes the 64×64 velocity at t = 5 by at most 4.3e-06.



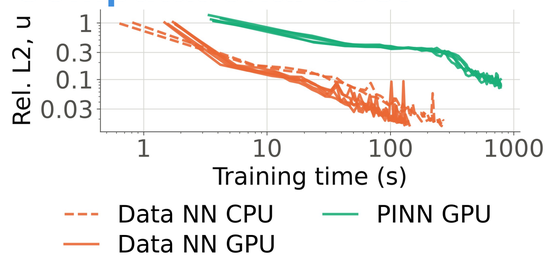




Accuracy & Physics



Computational Cost



	Train (s)	To field (s)	Infer. (ms)	RAM (GB)	VRAM (MB)
FD 64² CPU	–	3.9	–	0.08	–
Data NN GPU	137	141	10.3	1.51	122
PINN GPU	782	782	10.2	1.59	708
Data NN CPU	269	272	66	1.07	–
PINN CPU †	0.30 s/step	–	–	1.85	–

To field: FD = solve; data NN = FD solve + training; PINN = training. Infer.: full 64×64×51 field, median of 20 calls. RAM, VRAM: peak. † 2,000-step timing run.

Discussion

Accuracy. The data NN is closer to FD: test-A relative L2 of u 0.017 ± 0.001 vs 0.082 ± 0.011 (PINN; mean ± std, 5 GPU seeds). Test B is similar: 0.014 vs 0.077.

Physics. PINN fields obey the equations more closely: RMS continuity residual 0.071 vs 0.55, x-momentum 0.096 vs 5.6 (data NN).

Cost. GPU training: PINN 782 s, data NN 137 s; one FD solve 3.9 s (CPU). Training is repaid after about 36 (data NN) or 200 (PINN) full-field evaluations instead of FD runs.

Conclusion

At Re = 100 the PINN learned the cavity flow without any simulation data, and its fields obey the Navier–Stokes equations more closely than the data-driven network's. The data-driven network is more accurate against FD and trains faster. For a single field, FD is cheapest. Next: Re 400 and PINN sensitivity to collocation and loss weighting.

References

[1] Hou, S., Zou, Q., Chen, S., Doolen, G., and Cogley, A. C. 1995. Simulation of cavity flow by the lattice Boltzmann method. J. Comput. Phys. 118, 2, 329–347.

[2] Marchi, C. H., Suero, R., and Araki, L. K. 2009. The lid-driven square cavity flow: numerical solution with a 1024 x 1024 grid. J. Braz. Soc. Mech. Sci. Eng. 31, 3, 186–198.

[3] Raissi, M., Perdikaris, P., and Karniadakis, G. E. 2019. Physics-informed neural networks: a deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations. J. Comput. Phys. 378, 686–707.

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